MIL-P-20486A(ORD)
21 May 1953
SUPERSEDING
MIL-P-20486
3 December 1951

MILITARY SPECIFICATION

PRIMER ELEMENTS, DETONATORS
AND RELAYS
METAL PARTS FOR

1. SCOPE

1.1 This specification covers metal parts for primer elements, detonators, and relays.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and all Ordnance Corps drawings and Standard Inspection Procedures pertaining to the primer elements, detonator, or relay metal parts under contract and all drawings and specifications referenced thereon, of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS
MILITARY

MIL-G-2550	-	General Specification For Ammunition Except Small Arms Ammunition.
MIL-P-10025	-	Packing and Marking for Domestic Shipment of Inert Ammunition Components, General Specification.

(Copies of drawings, specifications, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 The metal parts shall comply with all requirements specified on applicable drawings and specifications.

3.3 The open end of the cups and retainers shall be trimmed by a method which shall result in an edge, with no burrs on either the outside or inside.

3.4 The base of the detonator cups, and primer cups where required, shall be flat, and the radius on the corner shall not exceed the dimensions specified on the drawing.

3.5 Cleaning of components. - The metal parts shall be thoroughly cleaned by a process which will leave them free of chips, dirt, grease, drawing compound, and other foreign material. The exterior and interior surfaces of the metal parts shall be free of stains and other evidence of attack by the cleaning agents used. Immediately following the cleaning operation, the parts shall be rinsed in such a manner as to prevent the drying of the cleaning fluid and the resultant formation of a powdered deposit which is very difficult to remove and cause difficulty in the loading operations. This rinse shall be followed by

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:55Z
Check the source to verify that this is the current version before use.

3.6 Brightening of components (applicable to copper and copper alloy parts only).-After cleaning, the copper and copper alloy cups shall be treated by dipping the metal parts in a solution consisting of 10 percent commercial sulphuric acid and then rinsing in cold water. Remove and then rinse in hot water and dry.

3.7 Workmanship.

3.7.1 All parts shall be free of chips, dirt, grease, and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents used.

3.7.2 The metal parts shall be smooth, bright, and free of all dents, laps and folds. Heat treatment of copper and copper alloy parts, if required by the drawing, shall be done after the cleaning operation specified in 3.5. If, following the heat treatment the parts are not bright, they shall be brightened by the method specified in 3.6 and in such a manner that cold working of the components shall not take place.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.-A lot shall consist of all components made by one manufacturer in one unchanged process in accordance with the same drawing, drawing revision, same specification and specification revision.

4.2 Sampling.-Number of samples, acceptance and rejection criteria used for determination of lot acceptance for test as specified herein shall be in accordance with the applicable Standard Inspection Procedure.

4.3 Inspection.-Inspection shall be as specified in Specification MIL-G-2550.

4.4 Crush test of primer cup (when required by applicable drawing).-Sample primer cups shall be placed on the bed of a press as shown in Figure 1. A load shall be applied in the axial direction so that the rate of travel of the testing machine shall not exceed .100 inch to .125 inch per minute, until the primer cup crushes as indicated by a falling off of the load.

5. PREPARATION FOR DELIVERY

5.1 Packaging, labeling and marking.-Packing, labeling and marking shall be as specified in Specification MIL-P-10025.

5.2 Data card.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 As primer, detonator, and relay parts are to be used in conjunction with the loading of high explosives, it is of the utmost importance that care be exercised in cleaning these components.

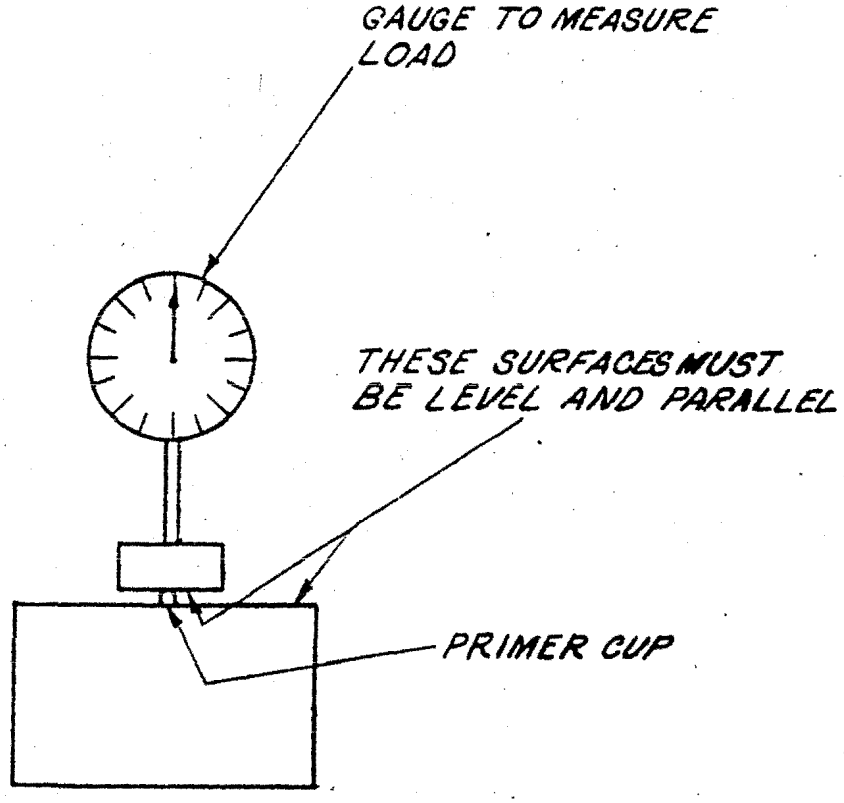
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:55Z
Check the source to verify that this is the current version before use.

Notice.-This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions is intended for use only in connection with procurement by the United States Government and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

Custodian:

Army-Ordnance Corps

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:55Z
Check the source to verify that this is the current version before use.



TEST FIXTURE FOR CRUSH TEST OF PRIMER CUP (SCHEMATIC)

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:55Z
Check the source to verify that this is the current version before use.